AML7001 Applied Machine Learning
Coursework 1: Model Evaluation Report

School of Computing, Northbridge University (2026-27, Teaching Block 1)

Key information

Weighting: 40% of the module mark.
Submission deadline: Thursday 12 November 2026, 14:00 (UK time).
Submission: one PDF report through the Coursework 1 submission point on the module page.
Length: 2,500 words, excluding references, tables, figures and the appendix.
Module leader: Dr Imogen Hale (i.hale@northbridge.example).

Assessment overview

This coursework assesses your ability to evaluate supervised learning models in a setting where the most obvious measure of success is misleading. It contributes to learning outcomes 2 (evaluate and compare predictive models using an appropriate methodology) and 4 (communicate technical findings to a non-specialist audience) in the module descriptor. The emphasis is on the quality of your reasoning about evaluation, not on obtaining the highest possible score from any one algorithm.

Scenario

Northbridge Community Health runs a network of outpatient clinics across the region. A substantial number of booked appointments are missed without notice, which wastes clinical time and lengthens waiting lists for other patients. The network is considering a reminder service in which staff telephone the patients judged most likely to miss their appointment. Staff time is limited, so only a small fraction of patients can be called each week, and the operations team has asked whether a predictive model could help decide whom to call.

You will use the Northbridge Clinic Appointments dataset (synthetic, 48,000 rows), available from the Coursework 1 folder. Each row is one booked appointment, described by the patient's age band, the lead time between booking and appointment, the clinic, the day of the week, previous attendance and whether an SMS confirmation was sent. The target column no_show is 1 when the patient did not attend.
Only about 7% of appointments in the dataset are missed.

What the report is for

Working in Python, you will prepare the data, train classification models and assess them as candidates for prioritising reminder calls. The report is the assessed output; your code supports it but is not marked separately.

Your report should open by framing the prediction problem in your own terms and describing the supplied data in enough detail for a reader to follow the decisions that come later. Write this opening for a reader from the operations team rather than for a machine learning specialist.

Particular attention should be paid to the class imbalance in the outcome: because missed appointments are comparatively rare, the report should consider how this affects the way the models are trained as well as the way their performance is judged. The evaluation measures you report should be chosen deliberately and justified by the practical cost of each kind of error in this setting; a model defended on its accuracy alone will not be regarded as adequately evaluated. Marks in this part of the report reward explanation more than the number of measures reported.

Your discussion should compare at least two different modelling approaches, with final performance reported on a held-out test set that played no part in model selection or tuning. Cross-validation on the training data is a sensible way to choose between settings, but its results are not a substitute for the final held-out comparison.

The report should close with a critical reflection on the limitations of your evaluation and on the further work you would recommend before any model was used to select patients for reminder calls. In that reflection, consider whether any choice made during training could have leaked information into the test results. Any resampling or class weighting must be applied to the training data only.

Presentation and submission

Most strong reports follow a structure close to this one, although it is not prescribed: introduction and problem framing, data, methods, results, discussion and limitations, conclusion. Put your code in an appendix or behind a link to a private repository. Name the file AML7001_CW1_<student number>.pdf and upload it to the Coursework 1 submission point before the deadline above. Late submissions are handled under the standard university rules described in the module handbook.

How the report is marked

Reports are marked against the School's postgraduate marking scheme. Work at distinction level (70% and above) is precise about what its results do and do not show, ties each evaluation decision to the clinic network's situation and is candid about uncertainty. Work at pass level (50-59%) applies suitable methods but explains them mechanically. Presentation and referencing account for 5% of the mark.

Academic integrity and support

This is an individual assessment. Generative AI tools may be used for help with code and language if their use is acknowledged in a short statement at the end of the report, in line with the School's policy. Lecture 7 and Workshop 7 cover the material this coursework draws on. Post questions about the brief on the module discussion board.